Ergodicity

When the average of many outcomes is the truth about one — and when believing so is fatal.

Contents

01 Two ways to average

02 Ergodic by nature

03 Ergodic by design

04 The pivot — a coin that multiplies

05 Non-ergodic in markets

06 Non-ergodic beyond markets

07 What remains

01

Two ways to average



A thousand coins, or one coin

- Flip a thousand coins once each. Average the outcomes across the thousand — that is the **ensemble average**.
- Flip one coin a thousand times. Average the outcomes along that one history — that is the **time average**.
- For a fair coin, both numbers land at one half.

Ergodic: the two averages agree

ensemble average = time average

A system is ergodic when the average taken across many independent instances equals the average taken along one instance, over time. It holds when nothing that happens on one draw can prevent the next draw from happening.

Example — a thousand coins

MEASURE	VALUE
Frequency of heads, across the thousand	0.501
Frequency of heads, along the one	0.497

Source: Ergodicity is a property, not a courtesy — for a coin, it happens to be true.

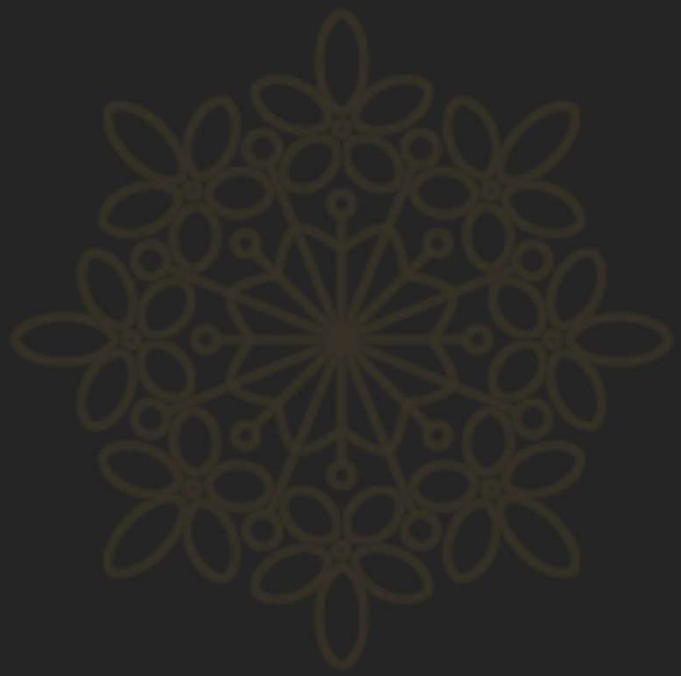
Example — three and a half

Measure	Value
Expected pip, one die	3.500
Mean, after 600 rolls of one die	3.492

Source: Under these conditions, and only these, the long run is a fair description of the short one.

02

Ergodic by nature



Boltzmann's box

- Roughly 10^{23} molecules in a gas — an ensemble no one could ever enumerate.
- Boltzmann's move: assume one molecule's history, watched long enough, samples the whole space of states.
- One history = one snapshot. The average over the room predicts the life of the particle.

The house

MEASURE	VALUE
House edge	2.70%
Bets per year	10,400,000
Largest single bet ÷ capital	0.0004

Source: The casino does not gamble. It has converted a risky wager into an arithmetic certainty.

Forty thousand policies

Measure	Value
Loss ratio	0.61
Largest claim ÷ capital	0.002
Correlation between claims, ρ	0.04

Source: Independence, and small claims relative to capital, are not accidents of the business — they are engineered.

The queue, and the vaccine

The Tuesday queue

Mean wait 4.20 minutes, 38 agents staffed. Arrivals are random, the mean wait is not — no queue on Tuesday can prevent Wednesday from happening.

Vaccine efficacy

94.1% efficacy, trial arm of 21,720. A population statistic that genuinely governs a population — the ensemble is the thing being protected.

QUESTION 1

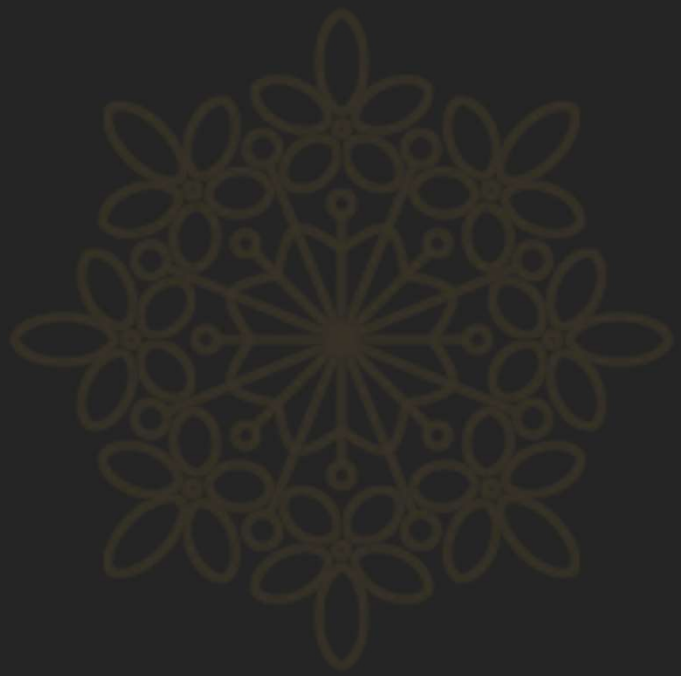
“What do the casino, the insurer, and Boltzmann's molecule actually have in common?”

Two conditions, and only two

- The draws are close to independent of one another.
- No single draw is large enough, relative to the whole, to change the outcome.
- Where both hold, the average taken across many is a fair description of one — and averaging is safe.

03

Ergodic by design



Bet sizing — the whole lesson, in advance

MEASURE	VALUE
Edge per hand	+0.60%
Stake ÷ bankroll	1.20%
Chance of ruin	0.3%

Source: A positive edge is not enough. The card counter sizes each stake small enough that no hand can end the sequence — ergodicity is manufactured by the bet size, not discovered in the cards.

Thirty mandates

Measure	Value
Books run	30
Worst book	−19%
Aggregate	+34%

Source: Thirty mandates, twenty years — his record is an average genuinely taken across portfolios.

Forty bets

Measure	Value
Written off	31 of 40
Returned the fund	2
Fund multiple	3.1×

Source: For the fund, the average of forty draws is an honest description of what the fund experiences.

03 · ERGODIC BY DESIGN

Ergodicity can be manufactured.

Small stakes and many independent mandates are how you build it.

04

The pivot — a coin that multiplies

×1.50 heads · ×0.60 tails

- Every round, a fair coin either multiplies your wealth by 1.50, or by 0.60.
- Expected value of one round: 0.5 × 1.50 + 0.5 × 0.60 = 1.05 — a five per cent expected gain.
- Ask the room: would you play this, repeatedly, with everything you have?

QUESTION 2

“The expected value of one round is +5%. Would you play sixty rounds, betting everything each time?”

Both statements are true

MEASURE	AFTER 60 ROUNDS
Growth rate along a typical path	−5.1%
Paths under 1.0 after 60 rounds	82%
Median path after 60 rounds	×0.04

Source: The ensemble average compounds at +5% a round forever. Almost every individual path goes to nothing. Only one of these two statements happens to you.

One Bernoulli, both averages

$$X = 1.50^{B} \cdot 0.60^{1-B}, \quad B \sim \text{Bernoulli}(\tfrac{1}{2})$$

$$W_{60} = 1.50^{K} \cdot 0.60^{60-K}, \quad K \sim \text{Binomial}(60, \tfrac{1}{2})$$

- Payoffs multiply, so only the *count* of heads matters — never the order.
- Both averages are now expectations of the same Bernoulli variable: one of X, one of ln X.

Four figures, one binomial

QUANTITY	FROM THE BERNOULLI	VALUE
Ensemble average	$E[X]^{60} = 1.05^{60}$	×18.7
Time-average growth	$E[\ln X] = \tfrac{1}{2} \ln 1.5 + \tfrac{1}{2} \ln 0.6$	−5.1% / round
Median path	$K = 30$, so 0.90^{30}	×0.042
Heads to break even	$K \geq 60 \ln(1/0.6) \div \ln 2.5$	34 of 60

Source: You need 56.7% heads merely to stand still; a fair coin pays 50%. That gap is the whole of the non-ergodicity.

04 · THE PIVOT

×18.7 vs ×0.04

The ensemble average after sixty rounds, against the median path — the same coin, sixty rounds, two entirely different fates.

The mechanism: an absorbing barrier

- Somewhere there is a level at which the sequence stops: bankruptcy, redemption, a margin call, a death.
- Above the barrier, you can average across draws as if they were independent trials.
- Touch it once, and there is no next draw. Every expected value computed behind you now describes a world you are no longer in.

04 · THE PIVOT

Ninety paths, forty rounds.

One of them is yours — and it only has to touch the floor once.

05

Non-ergodic in markets



Six men, or six pulls

MEASURE	VALUE
Prize	£5.0m
Chance of ruin, per pull	1 in 6
Expected value	+£4.17m

Source: Six men taking one pull each is a survivable statistic. One man taking six pulls is not the same situation — and no expected-value calculation can tell them apart.

The martingale

Measure	Value
Wins before the run	26
Bet by the eighth loss	£256
Bankroll	£125
Outcome	£0

Source: The average outcome was never available to this player. Only the sequence was.

Ten thousand simulations

MEASURE	VALUE
Median simulation	+26%
5th percentile	−31%
The path that happened	−34%
Months to closure	9

Source: The distribution was correct. The fund still died, because the fund got one path — and the redemption gate closed at the bottom.

The manager and the marquee

Thirty mandates

An ensemble: his worst mandate is a footnote, his career survives every one of them. Leverage 1.0×, careers ended: 0.

One flagship fund

One levered path, in public, with redemption rights attached. Leverage 4.5×, careers ended: 1 — at the margin call.

Twenty-five to one

MEASURE	VALUE
Equity, four good years	×2.9
Balance-sheet leverage	25 : 1
Loss modelled as	a 1-in-10^9 event
Equity, five weeks	−92%

Source: Long-Term Capital Management, 1994–98. The event was rare in the ensemble. It was certain, eventually, on the path.

Ninety-nine per cent confident

Measure	Value
Confidence	99%
Breaches expected	1 day in 100
Trading days, 40 years	10,000
The breach, when it came	−£4.9bn

Source: Value-at-risk. Over forty years, that is a hundred breaches — and the model says nothing about their size.

Correlation eats the ensemble

MEASURE	VALUE
Contracts written	≈ 400,000
Assumed correlation, ρ	0.04
Realised correlation, ρ	0.97
Effective number of risks	1

Source: AIG's credit-default-swap book, 2008. The correlation changed, and the ensemble collapsed into a single path.

The fund's forty, the founder's one

MEASURE	VALUE
Fund outcome	3.1×
Companies failed	31 of 40
Founder outcome	0
Founder's draws	1

Source: The fund's arithmetic is honest across forty draws. The founder has one draw, seven years of it, and no diversification available to him.

QUESTION 3

“The fund's statistic describes the fund fairly. Does it describe the founder at all?”

Same returns, wrong order

MEASURE	VALUE
Average annual return, both retirees	6.5%
Drawn each year	7%
Bad years first	ruin at age 78
Bad years last (same years, reversed)	0.99× capital at age 80

Source: Sequence-of-returns risk. Path dependence in its purest form — no probability, only order.

06

Non-ergodic beyond markets



A life is not a portfolio

- The same four shocks — to health, career, reputation — struck in two different orders.
- D → C → B → A, heavy shocks first while there is vigour: the person reaches 85.
- A → B → C → D, heavy shocks last, capacity to recover already exhausted: perishes at 71.

Source: A life is worse than a portfolio — the capacity to absorb a shock declines with age, exactly when the shocks tend to get larger.

One in ten thousand years

MEASURE	VALUE
Design basis, sea wall	5.7 m
Annual probability of exceedance	0.01%
Observed wave height	14.5 m
Plants available	1

Source: Fukushima Daiichi, 2011. The wall was built to a number describing ten thousand years of coastline — the plant only ever had one of them.

Below the coldest datum

MEASURE	VALUE
Coldest prior launch	53 °F
Launch morning	31 °F
Flights flown	24
Flights available	1

Source: Challenger, 1986. Twenty-four flights had flown and the average flight was fine — but the average flight was never the question.

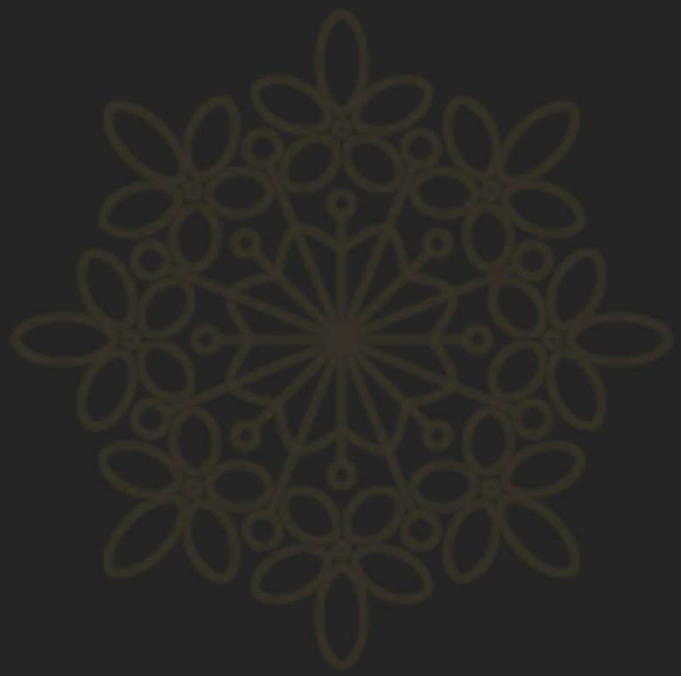
One Earth, no control group

MEASURE	VALUE
Ensemble members (model runs)	80
Median warming, 2100	+2.4 °C
Upper tail	+4.1 °C
Planets available	1

Source: If the path we are on is a tail run, the median was never a forecast about us.

07

What remains



07 · WHAT REMAINS

You do not receive the average. You receive the path you are on.

And it only has to end once.

Five questions before you average

- **Is there a barrier?** A level — ruin, redemption, margin call — that ends the sequence.
- **Are the draws independent?** Or does one bad outcome make the next one more likely?
- **Is any single draw large enough to matter?** Relative to capital, to career, to a life.
- **Whose ensemble is it?** A fund's average may not describe the founder, or the client, at all.
- **Can you manufacture ergodicity?** Smaller stakes, more independent bets, more time between draws.

Ergodic vs non-ergodic, side by side

	ERGODIC	NON-ERGODIC
Payoffs	Additive	Multiplicative, or barrier-bound
Draws	Independent, small relative to capital	Correlated, or large enough to end the game
The average	Describes one instance	Describes an ensemble you are not in
Safe response	Size up, diversify, repeat	Size down, avoid the barrier, survive first

07 · WHAT REMAINS

“Path dependence does not refine the average. It replaces it.”

— for anyone who does not get to live in the ensemble

Discussion

“Which of your own investing habits are quietly assuming an ensemble you don't actually get to live in?”